SOFTWARE REQUIREMENTS SPECIFICATION

Brenda's Poultry Farm — Egg Production Tracking System

Document	*Details*
Version	1.0
Date	20 October 2026
Prepared by	Group A Members
Client / organisation	Brenda's Poultry Farm, Muyenga

Version history

Version	*Date*	*Change*	*By*
1.0	20 October 2026	First SRS draft based on interview and observation notes.	Project group

1. Introduction

1.1 Purpose

This document describes the requirements for a small Egg Production Tracking System for Brenda's Poultry Farm in Muyenga. It will guide the project group when planning and designing the system and help the farm check whether it meets its main need.

1.2 Scope

The smartphone-only system will record egg production and reduce mistakes caused by manual calculations in the counter book.

In scope	Out of scope in this version
Record date and number of eggs collected.	Record sales or payments.
Record damaged eggs separately.	Manage general stock or inventory.
Calculate daily and weekly production totals.	Manage flock health or treatment.
View saved production records and summaries.	Provide a desktop/computer version.

1.3 Definitions

Term	Meaning
Production record	Eggs collected and recorded for a particular date.
Damaged eggs	Eggs with cracks or other damage, recorded separately.
Daily total	Number of eggs recorded for one day.
Weekly total	Eggs recorded over a seven-day period.

1.4 References

Questionnaire interview notes for Brenda's Poultry Farm (19 October 2026); observation of two egg-production counter books (19 October 2026); PCSE211 Software Engineering Principles, Unit 4, Section 4.8 sample SRS.

2. Overall description

2.1 Product perspective

This proposed system will support the farm's current paper-based records. It will save egg-production entries and calculate totals automatically. It will not manage other farm activities.

2.2 Users

USER	MAIN TASKS	EXPECTED SKILLS	APPROXIMATE NUMBER
POULTRY WORKERS	Collect eggs from the hens' houses, sort eggs, identify damaged eggs and count the eggs collected each day.	Basic poultry farming and counting skills	3
BRENDA (OWNER)	Review production records, check the daily and weekly totals, and confirm that the figures are correct.	Basic smartphone skills	1
RECORD-KEEPING WORKER	Enter daily egg production figures into the smartphone system and check that the entries are correct.	Basic smartphone skills	1

2.3 Operating environment

The system will be used on a smartphone. Screens should fit the phone display and use clear labels and simple forms.

2.4 Constraints

The first version must stay small and focus on egg production tracking. The farm has about 70 laying hens and produces around 50–60 eggs per day. No computer version is planned.

2.5 Assumptions and dependencies

A worker will enter daily figures and Brenda will check and confirm totals. The system depends on users entering records regularly and correctly. Technical details and backup arrangements will be decided during later design.

3. Functional requirements

Priority guide: Must = essential to the main purpose; Should = important but not essential for the first usable version; Could = useful extra; Won't this time = excluded from this version.

ID	Requirement	Priority	Source
FR-01	The system shall let a worker enter the date and number of eggs collected each day.	Must	Interview
FR-02	The system shall let the user record damaged eggs separately from intact eggs.	Must	Interview
FR-03	The system shall save each production entry for later viewing.	Must	Observation; project scope
FR-04	The system shall calculate the total eggs recorded for a selected day.	Must	Interview: manual calculation errors
FR-05	The system shall calculate the total eggs recorded for a selected week.	Must	Interview: need for weekly figures
FR-06	The system shall show intact and damaged egg counts separately for a selected day.	Must	Interview
FR-07	The system shall let Brenda review and confirm saved production entries.	Should	Interview: owner checks totals
FR-08	The system shall let a user view previous production records by date.	Should	Observation; project scope
FR-09	The system shall warn the user if a required field, such as date or egg count, is missing.	Should	Project scope: reduce incomplete records
FR-10	The system shall let a user correct an entry when a recording mistake is found.	Should	Project scope: support accuracy

4. Non-functional requirements

The targets below are proposed and can be checked during testing.

ID	Type	SMART requirement	Priority
NFR-01	Performance	The system shall display a requested daily or weekly total within 3 seconds under normal conditions.	Must
NFR-02	Usability	A worker with basic smartphone skills shall enter and save a sample record correctly after no more than 30 minutes of training.	Must
NFR-03	Security	The system shall require an authorised user to sign in before viewing or changing saved records.	Must
NFR-04	Reliability	A saved production record shall still be available after the user closes and reopens the system.	Must
NFR-05	Compatibility	The main entry and summary screens shall fit a smartphone display without horizontal scrolling.	Should

5. External interface

User interface: a smartphone screen with simple fields for date, intact egg count and damaged egg count, plus clear buttons to save entries and view totals. The system should use plain English and readable text.

6. Acceptance criteria

Requirement	Test	Expected result
FR-02, FR-06	Enter 55 intact eggs and 3 damaged eggs for one date.	The two counts remain separate and display correctly.
FR-04, FR-05	Enter sample records for several days and request daily and weekly totals.	Totals match the sum of saved entries.
NFR-01	Request a daily or weekly total and time the result.	The total appears within 3 seconds.
NFR-04	Save a record, close the system and reopen it.	The saved record is still available.

7. MoSCoW and SMART justification

Recording production, separating damaged eggs, saving records and calculating totals are Must-have requirements because without them the system would not solve the main problem. Reviewing past records and correcting entries are Should-have requirements because they improve accuracy but come after the basic functions. Sales and flock-health features are excluded to keep this version focused. The non-functional requirements are measurable: for example, the total must appear within 3 seconds and a worker must be able to complete a sample entry after no more than 30 minutes of training.

8. Requirements-gathering evidence

Interview: the worker explained that daily production is written in a large counter book, manual calculations can be wrong, damaged eggs are counted separately, and Brenda checks and confirms totals. The farm has about 70 laying hens and produces around 50–60 eggs per day.

Observation: about two counter books containing egg-production records were observed. This supports the need for an easier way to save and find production figures. Any extra details about the books should only be added if they were actually seen.